Spellcast

Your room, within reach.

Gesture control for lights and music



Build Day / Durham / 17 October 2026

Concept illustration

Smart rooms still interrupt us

Students with smart desk setups interrupt study
to adjust lights and music through separate controls.

40%

of UK households had a smart speaker

3 in 5

smart-speaker users have had the wrong radio station play after a request

Sources: Ofcom, Media Nations 2025 and Audio Listening 2025. Demand for gestures remains a hypothesis.

Gesture control for lights and music

01 **Lighting**
Power, brightness and colour on the paired Zigbee lamp

02 **Music**
Spotify playback, volume and next track

03 **Room scenes**
Calm, Study and Party coordinate light and music

Current scope: macOS, webcam, calibrated targets, paired Zigbee lamp and the Spotify desktop app.

Watch the room respond

LIVE DEMO

Lamp

Point to select.
Open or close your hand.
Raise or lower brightness.

Music

Point at the speaker.
Raise or lower the
Spotify volume.

Scene

Tap index fingertips
together twice to change
light and music together.

Webcam perception runs on the Mac. Lamp control needs the local Zigbee setup. Spotify remains its own service.

A different way to choose the device

Approach	Device selection	Main requirement
Apps / dashboards	Choose an item on a screen	Phone or desktop interaction
Voice assistants	Speak a device name	Microphone and voice command
Flic Duo	Press or gesture with a controller	Dedicated physical control
Spellcast	Point at a calibrated object	Mac webcam and local device setup

Our bet: pointing feels more direct for visible desk devices.

Product sources: Philips Hue and Flic. Comparison reflects interaction methods, not a usability benchmark.

UK market and launch scope

TAM / category proxy

11.6m

UK households with a smart speaker

SAM / assumed fit

116k

1% assumed to fit the initial supported setup

SOM / year-one target

1,000

Paying rooms after a successful beta

£29,000 annual gross revenue at 1,000 paying rooms

TAM is a broad proxy. SAM, SOM and £29/year are planning assumptions. Sources: ONS and Ofcom, 2025 data.

The first 100 users

100

activated setups

01 Durham pilot

Recruit 20 compatible testers through computing and gaming societies.

02 A demo people can share

Publish the real interaction with a setup guide. Reach maker groups and desk-setup creators.

Target funnel: 400 qualified visits × 25% activation = 100 users. Proposed channels and conversion, not traction.

A software licence for the room

£29

per room / year

Free beta

Trial with compatible hardware before a paid launch.

Proposed paid offer

Supported lighting and music controls, with ongoing compatibility updates.

Pricing hypothesis. Hardware and music service costs are separate. No cloud video inference is required.

Spellcast

Your space, within reach.

20 students. One room each.

Join the first pilot.



Concept illustration

Prototype scope and boundaries

Implemented in the repo

Two-hand tracking and calibrated targeting
Lamp power, brightness and colour
Spotify controls and coordinated scenes
Local model assets and camera processing

Current limits

Mac must stay awake with camera in view
Targets need manual calibration
Lamp needs Zigbee2MQTT and a coordinator
Spotify needs macOS Automation access

Hardware demo is the evidence for physical performance. Other brands, arbitrary rooms and accessibility remain unvalidated.

Market assumptions and sensitivity

Scope	Room / household calculation	Annual gross at £29
TAM proxy	29m UK households × 40% = 11.6m	£336.4m theoretical
SAM scenario	11.6m × assumed 1% fit = 116k	£3.364m scenario
SOM target	1,000 paid rooms in year one	£29k target

At 0.5% fit: 58k rooms. At 2% fit: 232k rooms.

Broad category proxy, assumed compatibility and adoption, then a sales target. None are measured product demand.

The pilot tests everyday usefulness

Task reliability	At least 90% of prompted actions succeed across pilot tasks
Accidental control	No more than one unintended command per user-hour
Continued use	At least 10 of 20 testers choose to use it after one week
Payment signal	At least 5 of 20 make a paid reservation after the trial

Proposed go / no-go thresholds. Compare with current controls and measure camera comfort, fatigue and setup friction.

Research sources

01 **Ofcom / Media Nations 2025**

40% UK household smart-speaker ownership. Printed page 92.

02 **Ofcom / UK listening habits 2025**

Three in five users report incorrect radio-station playback.

03 **ONS / Families and households, 2025**

29.0 million UK households. Released 17 April 2026.

04 **Philips Hue and Flic**

Existing app, switch and physical gesture-control alternatives.

05 **Spellcast / current source code**

Features and current integration boundaries, commit 23970ca.

Full links and source qualifications appear in the relevant speaker notes. Cover visual is a concept illustration.